

Deployment Documentation

Specifications Overview

The Healthcare Predictive Analytics Platform uses a Laravel 10-based web application that allows users to input lifestyle and health factors to obtain a predicted risk level for Type 2 Diabetes (Low, Medium, High).

The system integrates machine learning, data preprocessing, dynamic forms, and real-time prediction using a Python model backend.

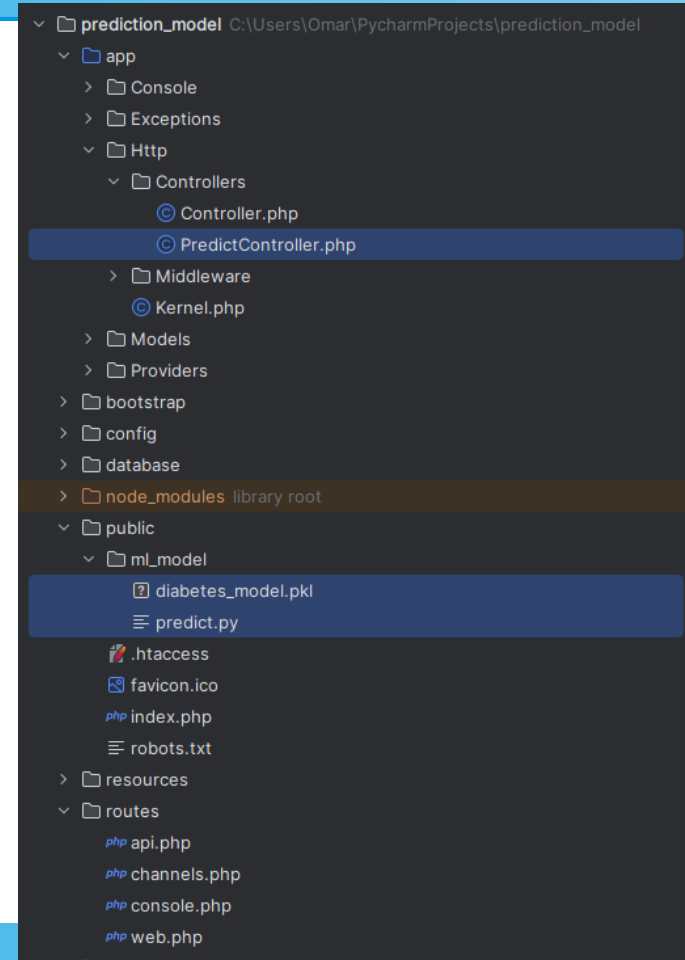
Main Layout

The application consists of 3 main entrypoints

The main controller: app/Http/Controllers/PredictController.php

The python script: Public/ml_model/Predict.py

The pkl file: public/ml_model/diabetes_model.pkl



The Laravel View

- Input fields that correlate to the 19 features

Diabetes Type 2 Risk Prediction

High Blood Pressure (doctor diagnosed)?

Yes

High Cholesterol?

Yes

Cholesterol checked last 5 years?

Yes

BMI

55

Smoker?

Yes

Ever had a stroke?

Yes

Heart disease or attack?

Yes

Exercised in last 30 days?

Yes

Diet habits

☐ None ☐ Fruits ☒ Veggies ☐ Both

Heavy alcohol consumption?

Yes

Cont.

Heavy alcohol consumption?

Yes

General Health (1=Excellent → 5=Poor)

Mental Health (bad days)

Physical Health (bad days)

Difficulty walking?

Yes

Age

22

Education

High school graduate

Income

Less than \$10k

Sex

Male

Predict Risk

Diabetes Risk Prediction Result

High risk of Diabetes

88.81%

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Debug Info:

```
Raw Output: {
  "prediction": "High risk of Diabetes",
  "risk": 0.8881242049585353,
  "debug": {
    "raw_input": {
      "HighBP": 1,
      "HighChol": 1,
      "HeartDiseaseorAttack": 1,
      "Stroke": 1,
      "BMI": 55,
      "GenHlth": 3,
      "MentHlth": 3,
      "PhysHlth": 3,
      "Diet": "veggies",
      "Age": 22,
      "Education": 4,
      "Income": 1,
      "Sex": 1,
      "Smoker": 1,
      "HvyAlcoholConsump": 1,
      "PhysActivity": 1,
      "DiffWalk": 1,
      "CholCheck": 1
    },
    "df_columns": [
      "HighBP",
      "HighChol",
      "HeartDiseaseorAttack",
      "Stroke",
      "BMI",
      "GenHlth",
      "MentHlth",
      "PhysHlth",
      "Diet",
      "Age",
      "Education",
      "Income",
      "Sex",
      "Smoker",
      "HvyAlcoholConsump",
      "PhysActivity",
      "DiffWalk",
      "CholCheck"
    ]
  }
}
```